

NGUYEN HOANG THACH

Backend Engineer (Intern / Fresher)

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SUMMARY

Backend developer with experience building web applications using Node.js and TypeScript. Interested in backend systems, AI-powered applications, and continuously improving software engineering skills through real-world projects.

EDUCATION

Ho Chi Minh City University of Technology and Engineering Ho Chi Minh City, Vietnam
Bachelor of Engineering in Information Technology (GPA: 3.5/4.0, Expected Graduation: 2027) Sep 2023 – Present
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Operating Systems, Computer Networks

PROJECTS

AI Interview Platform [GitHub](#)

- Architected and developed an AI-powered interview preparation platform supporting CV analysis, **ATS resume optimization**, and real-time mock interviews through text and voice interactions.
- Built a **CV-to-JD matching** and optimization engine that performs gap analysis, identifies missing qualifications, and generates ATS-optimized resume versions tailored to target job roles.
- Developed a real-time voice interview system integrating **Speech-to-Text (STT)**, LLM processing, and **Text-to-Speech (TTS)**, enabling natural AI-to-candidate conversations similar to virtual meeting platforms.
- Implemented **Server-Sent Events (SSE)** for real-time AI response streaming, reducing perceived latency and improving conversational experiences during interview sessions.
- Developed an **AI Evaluation Framework** based on rubric-driven assessment, leveraging **Structured LLM Outputs** and **Schema Validation** to generate consistent interview evaluations and actionable feedback; utilized **Redis** for caching and JWT token revocation, and **BullMQ** for asynchronous email delivery and user notification processing.
- Tech Stack:* TypeScript, Node.js, Express.js, MongoDB, Prisma ORM, Redis, BullMQ, JWT, SSE, OpenAI API, Google Gemini, Zod, TTS/STT APIs.

E-Learning Platform [Live Demo](#) · [GitHub](#)

- Developed the backend architecture for an online learning platform supporting course management, enrollments, payments, and asynchronous media processing.
- Built RESTful APIs using **NestJS** and **Prisma ORM**, enforcing strict request validation and end-to-end type safety with **Zod**.
- Architected an asynchronous video processing pipeline where clients upload media directly to **Cloudflare R2** via **Presigned URLs**, triggering **Celery** and **Redis** background jobs for HLS transcoding and AI-powered multilingual dubbing using **Whisper**, machine translation, and **TTS** services.
- Containerized **NestJS**, **PostgreSQL**, **Redis**, and **Celery** services using **Docker Compose** to ensure environment consistency and simplify deployment workflows.
- Secured the platform with **JWT Authentication** and **Role-Based Access Control (RBAC)** for user authorization and access management.
- Integrated **Cloudinary** for scalable image optimization and utilized **WebSockets** for real-time bidirectional user notifications.
- Tech Stack:* NestJS, TypeScript, Prisma ORM, PostgreSQL, Redis, Celery, Cloudflare R2, Cloudinary, Zod, WebSockets, JWT, Docker.

BlinkyVocab - Gamified Vocabulary Learning Platform [Live Demo](#) · [GitHub](#)

- Developed the backend architecture for a gamified vocabulary learning platform supporting flashcards, quizzes, progress tracking, and community-driven vocabulary sharing.
- Implemented a personalized learning engine combining **Spaced Repetition (SM-2)** and the **Forgetting Curve** to optimize review intervals and improve long-term retention.
- Integrated AI services to automatically generate vocabulary metadata, CEFR levels, examples, meanings, and learning content from user-provided words and phrases.
- Built content extraction and lesson generation features that transform articles and text passages into contextual vocabulary learning materials.
- Tech Stack:* NestJS, TypeScript, Prisma ORM, PostgreSQL, Redis, WebSockets, JWT, OpenAI API.

SKILLS

Programming Languages:	TypeScript, JavaScript, Java, SQL, Python
Backend Frameworks:	NestJS, Express.js
Databases:	PostgreSQL, MongoDB, MySQL
Technologies & Tools:	Redis, BullMQ, Docker, SSE, WebSocket, Elasticsearch, Prisma ORM, JWT, Git
Languages:	Vietnamese (Native), English (B1 – TOEIC 600)

ACTIVITIES & AWARDS

- Winner, Best Idea Award – HCMUTE Hackathon 2025** 2025
- Awarded for developing “TamTro” — an AI-powered mental health support platform.
 - Developed backend RESTful APIs using Node.js and Express.js for authentication, real-time chat, and emotion analysis features.